The Measurement Loop: A Complete Cycle in RLI

测量环：域内锁定推断中的完整测量周期

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Abstract.
The measurement loop is a closed curve on the S^1 double cover. The parameter θ_t runs from 4π down to 0, completing one full measurement cycle. The phase diagram spans ground state, physical-layer activation, inflation, black-hole formation, wormhole freeze, flip point, and white-hole eruption.

摘要.
测量环是 S^1 双覆盖上的一条闭合曲线。参数 θ_t 从 4π 减小到 0，完成一次完整测量周期。相图覆盖基态、物理层激活、暴胀、黑洞形成、虫洞冻结、翻转点、白洞喷发。

I. Definition / 定义

S^1 double cover $z \to z^2$. θ_t has period 4π.

S^1 双覆盖 $z \to z^2$。θ_t 周期 4π。

Measurement loop = a closed curve on the S^1 double cover. Parameter $\theta_t \in [0, 4\pi)$.

测量环 = S^1 双覆盖上的闭合曲线。参数 $\theta_t \in [0, 4\pi)$。

Coordinates $(\gamma_{bg}, \kappa_{AB}, \kappa_{BC}, \kappa_{CA})$. γ_{bg} and κ_{AB} are linearly dependent; effective dimension is 3.

坐标 $(\gamma_{bg}, \kappa_{AB}, \kappa_{BC}, \kappa_{CA})$。$\gamma_{bg}$ 与 κ_{AB} 线性相关，有效维度 3。

II. Amplitude / 幅度

$$\gamma_{bg}(\theta_t) = \Gamma \left(1 - \cos \frac{\theta_t}{2}\right)$$

Single peak. $\gamma_{bg} = 2\Gamma$ at $\theta_t = 2\pi$; $\gamma_{bg} = 0$ at $\theta_t = 0, 4\pi$.

单峰。$\theta_t = 2\pi$ 处 $\gamma_{bg} = 2\Gamma$ 最大；$\theta_t = 0, 4\pi$ 处 $\gamma_{bg} = 0$。

III. Direction and Chirality / 方向与手性

$$\kappa_{ij}(\theta_t) = \kappa_0 \left(1 - \cos\left(\frac{\theta_t}{2} + \phi_{ij}\right)\right)$$

$\phi_{AB} = 0,\ \phi_{BC} = \delta,\ \phi_{CA} = 2\delta$, with $\delta = 2/9$.

$\phi_{AB} = 0,\ \phi_{BC} = \delta,\ \phi_{CA} = 2\delta$，其中 $\delta = 2/9$。

Three peaks staggered by δ: κ_{AB} peaks at 2π, κ_{BC} at $2\pi - 2\delta$, κ_{CA} at $2\pi - 4\delta$.

三峰错开 δ：κ_{AB} 峰在 2π，κ_{BC} 峰在 $2\pi - 2\delta$，κ_{CA} 峰在 $2\pi - 4\delta$。

Chirality arises from the stagger. $\kappa_{AB} \neq \kappa_{BA}$.

手性来自三峰错开。$\kappa_{AB} \neq \kappa_{BA}$。

IV. Phase Diagram / 相图

Interval / 区间	θ_t	γ_{bg}/Γ	Stage / 阶段
A	$4\pi \to 3\pi$	0 → 1	Ground state → physical layer activation / 基态 → 物理层激活
B	$3\pi \to 2\pi$	1 → 2	Inflation / 暴胀
C	2π	2	Black-hole formation / 黑洞形成
D	$2\pi \to \pi$	2 → 1	Wormhole freeze / 虫洞冻结
E	π	1	Flip point (physical-layer resonance) / 翻转点（物理层共振点）
F	$\pi \to 0$	1 → 0	White-hole eruption / 白洞喷发
G	0	0	Return / 回归

Unification with layer lock.

The resonance condition $\gamma_{bg,i} = \Gamma_i$ requires $\theta_t/2 + \phi_i = \pi/2$, i.e. $\theta_t = \pi - 2\phi_i$. Physical layer $\phi = 0$: $\theta_t = \pi$. Biological layer $\phi = \delta$: $\theta_t = \pi - 2\delta$. Symbolic layer $\phi = 2\delta$: $\theta_t = \pi - 4\delta$. The "flip point" at $\theta_t = \pi$ in the phase diagram is the same structure as the physical-layer resonance.

与层锁的统一。

层锁共振条件 $\gamma_{bg,i} = \Gamma_i$ 要求 $\theta_t/2 + \phi_i = \pi/2$，即 $\theta_t = \pi - 2\phi_i$。物理层 $\phi = 0$：$\theta_t = \pi$。生物层 $\phi = \delta$：$\theta_t = \pi - 2\delta$。符号层 $\phi = 2\delta$：$\theta_t = \pi - 4\delta$。测量环相图中 $\theta_t = \pi$ 标“翻转点”，与物理层共振点是同一结构。

V. Hierarchy / 层级

推导

Loop definition, $\gamma_{bg}(\theta_t)$, $\kappa_{ij}(\theta_t)$, three-peak stagger, phase diagram structure.

测量环定义、$\gamma_{bg}(\theta_t)$、$\kappa_{ij}(\theta_t)$、三峰错开、相图结构。

候选

Structural asymmetry 0.774 (quantitative stagger result, formula not derived).

结构不对称 0.774（三峰错开的定量结果，公式未推）。

接口

Black-hole entropy $1/4$, white-hole threshold $1.5\Gamma_{3D}$, Hawking temperature. In another paper.

黑洞熵 $1/4$、白洞阈值 $1.5\Gamma_{3D}$、霍金温度。另文处理。